TrES-3b

R-0512

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_240720` · created 2026-07-20T20:31:16+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460511.7684 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.057
Transit depth (Rp/Rs)^2	3.82 +/- 2.22 [%]
Orbital Inclination (inc)	85.61 +/- 2.89
Airmass coefficient 1 (a1)	0.74 +/- 0.16
Airmass coefficient 2 (a2)	0.32 +/- 0.83
Scatter in the residuals of the lightcurve fit is	22.88 %
Best Comparison Star	#2 - [230, 69]
Optimal Aperture	2.05
Optimal Annulus	8.2
Transit Duration (day)	0.075 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.057 vs 0.16309 (deviation 0.56σ)
Duration fit vs published	0.075 d vs 0.0567 d (deviation 1.22σ)
Mid-time O–C	-5.0 min
Depth SNR	3.4
Residual scatter	22.88 %

Light curve

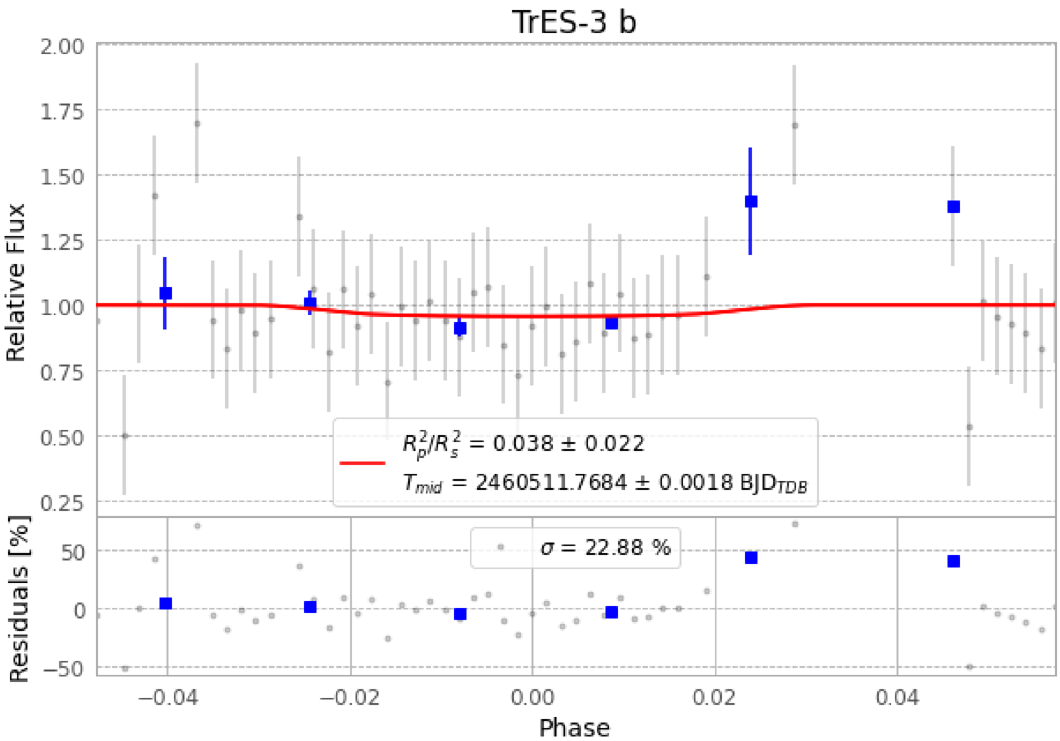

Figure 1 — FinalLightCurve_TrES-3 b_2024-07-19.png (SHA-256 681e59d06633f6db…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [302, 230], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e1d6ed9c7c9650be…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

68 FITS frames (45 MB), `TRES-3240720045115.FITS` → `TRES-3240720081215.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-240720.log (83a319717a3d…), FinalLightCurve_TrES-3 b_2024-07-19.png (681e59d06633…), FinalLightCurve_TrES-3 b_2024-07-19.csv (e52be32dd7c7…)

Compute resources

Wall time 245.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

